

T3 BSM terms grouped by field content

EFT order: 5; loop order: 1. Each hypercharge has a UV row and a matched BSM EFT row. Empty cells denote no term with that field combination.

Field combinations 1 of 4

Y	Lagrangian	$B_{\mu\nu}$	H	N	$W_{\mu\nu}$
-3	UV			$-\frac{1}{2}M_N N_{r_1}{}^T C N_{r_1}$	
-3	Matched	$\Delta\mathcal{L}_{\text{EFT}} - \frac{3\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}{}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}{}^\dagger \end{aligned} $		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$-\frac{5}{2}$	UV			$-\frac{1}{2}M_N N_{r_1}{}^T C N_{r_1}$	
$-\frac{5}{2}$	Matched	$\Delta\mathcal{L}_{\text{EFT}} - \frac{25\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}{}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}{}^\dagger \end{aligned} $		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
-2	UV			$-\frac{1}{2}M_N N_{r_1}{}^T C N_{r_1}$	
-2	Matched	$\Delta\mathcal{L}_{\text{EFT}} - \frac{2\hbar B_{\mu_1\mu_2}^2}{3\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}{}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}{}^\dagger \end{aligned} $		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$-\frac{3}{2}$	UV			$-\frac{1}{2}M_N N_{r_1}{}^T C N_{r_1}$	
$-\frac{3}{2}$	Matched	$\Delta\mathcal{L}_{\text{EFT}} - \frac{3\hbar B_{\mu_1\mu_2}^2}{8\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}{}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}{}^\dagger \end{aligned} $		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
-1	UV			$-\frac{1}{2}M_N N_{r_1}{}^T C N_{r_1}$	

Y	Lagrangian	$B_{\mu\nu}$	H	N	$W_{\mu\nu}$
-1	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{6\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}^\dagger \end{aligned} $	$-\frac{1}{2} M_N N_{r_1}^T C N_{r_1}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$-\frac{1}{2}$	UV				
$-\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + \frac{\lambda_3 \lambda_4 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}^\dagger \end{aligned} $	$-\frac{1}{2} M_N N_{r_1}^T C N_{r_1}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
0	UV				
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4 \lambda_3 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}^\dagger \end{aligned} $	$-\frac{1}{2} M_N N_{r_1}^T C N_{r_1}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
0	UV				
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$ \begin{aligned} & + \frac{\lambda_3^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4 \lambda_3 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & + \frac{\lambda_4^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & + C_{\text{H}2} H_{a_1} H_{a_1}^\dagger \\ & - \mu^2 H_{a_1} H_{a_1}^\dagger \end{aligned} $	$-\frac{1}{2} M_N N_{r_1}^T C N_{r_1}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$\frac{1}{2}$	UV				

Y	Lagrangian	$B_{\mu\nu}$	H	N	$W_{\mu\nu}$
$\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$ \begin{aligned} &+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger \\ &-H_{a_1}\mu^2H_{a_1}^\dagger \\ &+\frac{\hbar H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+\frac{\hbar H_{a_1}H_{a_2}\lambda_4^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ &+\frac{\hbar H_{a_1}H_{a_2}\lambda_3\lambda_4H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+\frac{\hbar\bar{\lambda}_5H_{a_1}H_{a_2}\lambda_5H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \end{aligned} $		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
1	UV			$-\frac{1}{2}M_NN_{r_1}{}^TCN_{r_1}$	
1	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{6\epsilon}$	$ \begin{aligned} &+\frac{\lambda_3^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+\frac{\lambda_4^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ &+\frac{\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger \\ &-\mu^2H_{a_1}H_{a_1}^\dagger \end{aligned} $		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$\frac{3}{2}$	UV			$-\frac{1}{2}M_NN_{r_1}{}^TCN_{r_1}$	
$\frac{3}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{8\epsilon}$	$ \begin{aligned} &+\frac{\lambda_3^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+\frac{\lambda_4^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ &+\frac{\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger \\ &-\mu^2H_{a_1}H_{a_1}^\dagger \end{aligned} $		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
2	UV			$-\frac{1}{2}M_NN_{r_1}{}^TCN_{r_1}$	
2	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{2\hbar B_{\mu_1\mu_2}^2}{3\epsilon}$	$ \begin{aligned} &+\frac{\lambda_3^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+\frac{\lambda_4^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ &+\frac{\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ &+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger \\ &-\mu^2H_{a_1}H_{a_1}^\dagger \end{aligned} $		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
$\frac{5}{2}$	UV			$-\frac{1}{2}M_NN_{r_1}{}^TCN_{r_1}$	

Y	Lagrangian	$B_{\mu\nu}$	H	N	$W_{\mu\nu}$
$\frac{5}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{25\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$+\frac{\lambda_3^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		
			$+\frac{\lambda_4^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
			$+\frac{\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		
			$+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger$		
			$-\mu^2H_{a_1}H_{a_1}^\dagger$		
3	UV			$-\frac{1}{2}M_NN_{r_1}^TCN_{r_1}$	
3	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+\frac{\lambda_3^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		
			$+\frac{\lambda_4^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$
			$+\frac{\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		
			$+C_{\text{H}2}H_{a_1}H_{a_1}^\dagger$		
			$-\mu^2H_{a_1}H_{a_1}^\dagger$		

Field combinations 2 of 4

Y	Lagrangian η	$H \eta$	$N D$	$\eta B_{\mu\nu}$
-3	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
-3	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
$-\frac{5}{2}$	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
$-\frac{5}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{25}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
-2	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
-2	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{2}{3}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
$-\frac{3}{2}$	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
$-\frac{3}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{3}{8}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
-1	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
-1	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$

Y	Lagrangian η	$H \eta$	$N D$	$\eta B_{\mu\nu}$
$-\frac{1}{2}$	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
$-\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
0	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_4\lambda_3\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		
0	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_4\lambda_3\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2\hbar H_{a_1}H_{a_2}\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}^\dagger H_{a_2}^\dagger$		
$\frac{1}{2}$	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\frac{1}{2}\overline{\lambda}_5 H_{a_1}H_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\frac{1}{2}\lambda_5\eta_{a_1}\eta_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$+\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$
$\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$+\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger$ $+\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3\lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)\overline{\lambda}_5 H_{a_1}H_{a_2}\lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$
1	UV	$+\frac{1}{2}\lambda_S\eta_{a_1}\eta_{a_2}\eta_{a_1}^\dagger\eta_{a_2}^\dagger$ $+M_\eta^2(-\eta_{a_1})\eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1}\eta_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$ $+\lambda_4\eta_{a_1}H_{a_2}H_{a_1}^\dagger\eta_{a_2}^\dagger$	$\frac{1}{2}iN_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1}(N_{r_1})$

Y	Lagrangian η	$H \eta$	$N D$	$\eta B_{\mu\nu}$
1	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{1}{6}\hbar \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) B_{\mu_1\mu_2}^2$
$\frac{3}{2}$	UV	$+\frac{1}{2}\lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$	$\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
$\frac{3}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{3}{8}\hbar \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) B_{\mu_1\mu_2}^2$
2	UV	$+\frac{1}{2}\lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$	$\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
2	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{2}{3}\hbar \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) B_{\mu_1\mu_2}^2$
$\frac{5}{2}$	UV	$+\frac{1}{2}\lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$	$\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
$\frac{5}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{25}{24}\hbar \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) B_{\mu_1\mu_2}^2$
3	UV	$+\frac{1}{2}\lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$ $+M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger$	$+\lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$ $+\lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$	$\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
3	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$\lambda_3^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\lambda_4^2 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$ $+\lambda_3\lambda_4 \hbar H_{a_1} H_{a_2} \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) H_{a_1}^\dagger H_{a_2}^\dagger$		$-\frac{3}{2}\hbar \log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right) B_{\mu_1\mu_2}^2$

Field combinations 3 of 4

Y	Lagrangian	ηD	$\eta W_{\mu\nu}$	$H \ell e$	$\eta N \ell$
-3	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
-3	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$-\frac{5}{2}$	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
$-\frac{5}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
-2	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
-2	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$-\frac{3}{2}$	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
$-\frac{3}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
-1	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
-1	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$-\frac{1}{2}$	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
$-\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
0	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
0	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
0	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$\frac{1}{2}$	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			$\eta_{a_2}\bar{\epsilon}_{a_1,a_2}\overline{y_{N,i_1,r_1}}N_{r_1}^T C P_L \ell_{a_1,i_1}$ $+\epsilon_{a_2,a_1}\eta_{a_1}^\dagger y_{N,i_1,r_1}\overline{\ell_{a_2,i_1}}P_R N_{r_1}$
$\frac{1}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8}\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{N,i_1,r_1}}Y_{e,i_3,i_2}\overline{\epsilon_{i_2}}P_L \ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8}\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}\epsilon_{i_2}}P_L \ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger$	
1	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
1	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$\frac{3}{2}$	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			
$\frac{3}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$		$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
2	UV	$+D_{\mu_1}(\eta_{a_1}) D_{\mu_1}(\eta_{a_1})^\dagger$			

Y	Lagrangian	$\eta\,W_{\mu\nu}$	$H\,\ell\,e$	$\eta\,N\,\ell$
2	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
$\frac{5}{2}$	UV	$+D_{\mu_1}\left(\eta_{a_1}\right)D_{\mu_1}\left(\eta_{a_1}\right)^\dagger$		
$\frac{5}{2}$	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		
3	UV	$+D_{\mu_1}\left(\eta_{a_1}\right)D_{\mu_1}\left(\eta_{a_1}\right)^\dagger$		
3	Matched $\Delta\mathcal{L}_{\text{EFT}}$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}^2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$		

Field combinations 4 of 4

Y	Lagrangian	$H \eta N \ell$	$H \eta N \ell e$
-3	UV		
-3	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
$-\frac{5}{2}$	UV		
$-\frac{5}{2}$	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
-2	UV		
-2	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
$-\frac{3}{2}$	UV		
$-\frac{3}{2}$	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
-1	UV		
-1	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
$-\frac{1}{2}$	UV		
$-\frac{1}{2}$	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
0	UV		
0	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
0	UV		
0	Matched $\Delta \mathcal{L}_{\text{EFT}}$		
$\frac{1}{2}$	UV		

[illegible]